Thus, for any fixed $c > 0$, strong model collapse occurs: the RHS vanishes only if $p_2 \to 0^+$, i.e only if we discard all but a vanishing proportion of synthetic data from the training dataset. This is strong model collapse. Combining with Corollary 1, we conclude that (at least in the isotropic setting, strong model collapse occurs both the under-parametrized and over-parametrized settings.

B.2 Connections to Classical Model Collapse in Regression

In the setting of classical model collapse (Shumailov et al., 2023, 2024; Alemohammad et al., 2023; Dohmatob et al., 2024b,a), we have $w_2^* = w_1^* + \sum_{\ell=1}^N X_\ell^\dagger E_\ell$,where N is the number of iterations (i.e self-loops) in the synthetic data-generation process. Let n_ℓ be the number of samples available for training at stage ℓ with training data $(X_\ell, X_\ell w_2^* + E_\ell) \in \mathbb{R}^{n\times d} \times \mathbb{R}^n$, where the noise vectors E_ℓ are independent with iid components from $N(0, \sigma_\ell^2)$. In the proportionate scaling regime $n_1, \ldots, n_N \to \infty$ with $d/n_\ell \to \phi_\ell \in (0,\infty)$ for all ℓ, the situation is equivalent to taking

$$\Delta = \sum_\ell \sigma_\ell^2 \cdot \mathbb{E}\,(X_\ell^\dagger)^\top X_\ell^\dagger \simeq \sum_\ell \frac{\sigma_\ell^2}{n_\ell - \mathrm{df}_2(\kappa_\ell;\Sigma)} \Sigma(\Sigma + \kappa_\ell I_d)^{-2}, \text{ with } \kappa_\ell = \kappa(n_\ell, 0; \Sigma).$$

In particular, if $\max_\ell \phi_\ell \le 1$ (so that there is enough samples to perfectly fit the training data at each stage of the iterative process), and for simplicity we set $\sigma_\ell = \sigma_0$ for all ℓ, then the above expression simplifies to $\Delta \simeq \left(\sum_\ell \sigma_\ell^2/(n_\ell - d)\right)\Sigma^{-1}$. More generally, consider the generic setting where $\Delta \simeq (c^2/d)\Sigma^{-1}$, for any $c > 0$, so that the previous setting corresponds to $c^2 = \sum_\ell \sigma_\ell^2 \phi_\ell/(1-\phi_\ell)$.

In the particular case where $p_1 \to 0^+$, i.e only synthetic data is available for training. Theorem 1 then gives

$$\zeta \simeq \frac{c^2}{d} \cdot \left(\mathrm{df}_2 + u\kappa^2 \operatorname{tr}(\Sigma + \kappa I_d)^{-2}\right) = \eta^2 \,\mathrm{df}_2 \cdot \left(1 + \kappa^2 \frac{\mathrm{df}_2}{n - \mathrm{df}_2} \operatorname{tr}(\Sigma + \kappa I_d)^{-2}\right).$$

In particular, taking $c^2 = \sum_\ell \sigma_\ell^2 \phi_\ell/(1-\phi_\ell)$ gives

$$\zeta \simeq \left(1 + \kappa^2 \frac{\mathrm{df}_2}{n - \mathrm{df}_2} \operatorname{tr}(\Sigma + \kappa I_d)^{-2}\right) \frac{\mathrm{df}_2}{d} \sum_\ell \frac{\sigma_\ell^2 \phi_\ell}{1 - \phi_\ell}. \tag{21}$$

This is recovers the main result of (Dohmatob et al., 2024a).

C Raw Bias-Variance Decomposition

Let X_j (resp. Y_j) be the design matrix (resp. response vector) corresponding to dataset $\mathcal{D}_j$. Thus, the design matrix $X_1 \in \mathbb{R}^{n_1 \times d}$ for the real dataset has rows given by x_i for $i \in [n_1]$ and $Y_1 \in \mathbb{R}^{n_1}$ with components y_i for $i \in [n_1]$, with $X_2 \in \mathbb{R}^{n_2\times d}$ and $Y_2 \in \mathbb{R}^{n_2}$ defined analogously for the synthetic dataset. Let $X \in \mathbb{R}^{n\times d}$(resp. $Y \in \mathbb{R}^n$) be the design matrix (resp. response vector) corresponding to the total dataset. We temporarily drop the condition $\Sigma_1 = \Sigma_2 = \Sigma$, and instead consider generally different covariance matrices Σ_1 and Σ_2 for the marginal distribution of the features x under the real data distribution P_1 and the synthetic data distribution P_2.

C.1 Classical Linear Model

Proposition 1. *Evaluated on the distribution $P_k = P_{\Sigma_k, \sigma_k^2, w_k^*}$, the test error of model $\widehat{f}_{CL}$ defined in (3) is given by*

$$E_{test}(\widehat{f}_{CL}) = B_k + V_k, \tag{22}$$

$$\text{where } V_k = \sum_{j=1}^2 \frac{\sigma_j^2}{n} r_j^{(4)}(I_d, \Sigma_k), \tag{23}$$

$$B_k = \begin{cases} r_2^{(3)}(\Delta, \Sigma_1) + \lambda^2 r^{(2)}(\Gamma, \Sigma_1), & \text{if } k = 1, \\ r_1^{(3)}(\Delta, \Sigma_2) + \lambda^2 r^{(2)}(\Gamma + \Delta, \Sigma_2) + 2\lambda r_1^{(4)}(\Delta, \Sigma_2), & \text{if } k = 2. \end{cases} \tag{24}$$

Proof. Indeed, one computes

$$\begin{aligned}
\mathbb{E}_{\mathcal{D}}\|\widehat{w} - w_k^*\|_{\Sigma_k}^2 &= \mathbb{E}_{X_1,Y_1,X_2,Y_2}\mathbb{E}_{x\sim N(0,\Sigma_k)}[(x^\top \widehat{w} - x^\top w_k^*)^2] \\
&= \mathbb{E}_{X_1,Y_1,X_2,Y_2}\, \|\widehat{w} - w_k^*\|_{\Sigma_k}^2 \\
&= \mathbb{E}_{X_1,Y_1,X_2,Y_2}\, \|(M+\lambda I_d)^{-1}X^\top Y/n - w_k^*\|_{\Sigma_k}^2 \\
&= \mathbb{E}_{X_1,Y_1,X_2,Y_2}\, \|(M+\lambda I_d)^{-1}X^\top (X_1 w_1^* + E_1, X_2 w_2^* + E_2)/n - w_k^*\|_{\Sigma_k}^2 \\
&= \mathbb{E}_{X_1,Y_1,X_2,Y_2}\, \|(M+\lambda I_d)^{-1}(M_1 w_1^* + M_2 w_2^*) - w_k^*\|_{\Sigma_k}^2 + V_1 + V_2 \\
&= B_k + V_{k,1} + V_{k,2}.
\end{aligned}$$

where

$$B_k := \mathbb{E}\,\|(M+\lambda I_d)^{-1}(M_k w_k^* + M_{-k} w_{-k}^*) - w_k^*\|_{\Sigma_k}^2, \tag{25}$$

$$V_{k,j} := \frac{\sigma_j^2}{n}\mathbb{E}\ \mathrm{tr}\, M_j (M+\lambda I_d)^{-1}\Sigma_k (M+\lambda I_d)^{-1} = \frac{\sigma_j^2}{n} r_j^{(4)}(I_d, \Sigma_k). \tag{26}$$

It remains to analyze the bias term B_k. To this end, observe that

$$(M+\lambda I_d)^{-1}M_k = I_d - (M+\lambda I_d)^{-1}(M_{-k}+\lambda I_d) = I_d - (M+\lambda M)^{-1}M_{-k} - \lambda (M+\lambda I_d)^{-1}.$$

Denoting $M_{-1} = M_2$, $M_{-2} = M_1$, $w_{-1}^* = w_2^*$, $w_{-2}^* = w_1^*$, and $\delta_k = (-1)^k\delta$, where $\delta := w_2^* - w_1^*$, we deduce that

$$\begin{aligned}
&(M+\lambda I_d)^{-1}M_k w_k^* + (M+\lambda I_d)^{-1}M_{-k}w_{-k}^* - w_k^* \\
&= (M+\lambda I_d)^{-1}M_{-k}w_{-k}^* - (M+\lambda I_d)^{-1}M_{-k}w_k^* - \lambda (M+\lambda I_d)^{-1}w_k^* \\
&= -(M+\lambda I_d)^{-1}M_{-k}\delta_k - \lambda (M+\lambda I_d)^{-1}w_k^\star.
\end{aligned}$$

Since w_1^* and $\delta_1 = \delta := w_2^* - w_1^*$are independent with distributions $N(0,\Gamma)$ and $N(0,\Delta)$ respectively, we deduce that

$$\begin{aligned}
B_1 &= \|(M+\lambda I_d)^{-1}M_2\delta - \lambda (M+\lambda I_d)^{-1}w_1^\star\|_{\Sigma_1}^2 \\
&= \mathrm{tr}\,\Delta M_2 (M+\lambda I_d)^{-1}\Sigma_1 (M+\lambda I_d)^{-1}M_2 + \lambda^2\,\mathrm{tr}\,\Gamma_1 (M+\lambda I_d)^{-1}\Sigma_1 (M+\lambda I_d)^{-1} \\
&= r_2^{(3)}(\Delta,\Sigma_1) + \lambda^2 r^{(2)}(\Gamma,\Sigma_1).
\end{aligned}$$

On the other hand, we have $B_2 = B_{2,1} + B_{2,2}$, where

$$\begin{aligned}
B_2 &= \| - (M+\lambda I_d)^{-1}M_1\delta - \lambda (M+\lambda I_d)^{-1}w_2^\star\|_{\Sigma_2}^2 \\
&= \| - (M+\lambda I_d)^{-1}M_1\delta - \lambda (M+\lambda I_d)^{-1}(w_1^\star + \delta)\|_{\Sigma_2}^2 \\
&= \| - (M+\lambda I_d)^{-1}\,(M_1 + \lambda I_d)\,\delta - \lambda (M+\lambda I_d)^{-1}w_1^\star\|_{\Sigma_2}^2 \\
&= \mathrm{tr}\,\Delta (M_1+\lambda I_d)(M+\lambda I_d)^{-1}\Sigma_2 (M+\lambda I_d)^{-1}(M_1+\lambda I_d) + \lambda^2\,\mathrm{tr}\,\Gamma (M+\lambda I_d)^{-1}\Sigma_2 (M+\lambda I_d)^{-1} \\
&= \mathrm{tr}\,\Delta M_1 (M+\lambda I_d)^{-1}\Sigma_2 (M+\lambda I_d)^{-1}M_1 + \lambda^2\,\mathrm{tr}\,\Delta (M+\lambda I_d)^{-1}\Sigma_2 (M+\lambda I_d)^{-1} \\
&\quad + 2\lambda\,\mathrm{tr}\,\Delta M_1 (M+\lambda I_d)^{-1}\Sigma_2 (M+\lambda I_d)^{-1} + \lambda^2\,\mathrm{tr}\,\Gamma (M+\lambda I_d)^{-1}\Sigma_2 (M+\lambda I_d)^{-1} \\
&= r_1^{(3)}(\Delta,\Sigma_2) + \lambda^2 r^{(2)}(\Gamma+\Delta,\Sigma_2) + 2\lambda r_1^{(4)}(\Delta,\Sigma_2).
\end{aligned}$$

This completes the proof. ☐

C.2 Random Projections Model

We now expand the test error $E_{test}(\widehat{f}_{RP})$ of the random projections model $\widehat{f}_{RP}$ defined in (5). For convenience, we recall the definition of the model here. Let S be a $d\times m$ random matrix with iid entries from $N(0,1/d)$. The model $\widehat{f}_{RP}$ is defined by $\widehat{f}_{RP}(x) := \Phi(x)^\top \widehat{v}$, where $\Phi(x) := S^\top x \in \mathbb{R}^m$ defines a random feature map, and $\widehat{v}\in\mathbb{R}^m$ is given by

$$\arg\min_{v\in\mathbb{R}^m} L(w) = \sum_k \frac{\|\Phi(X_k)v - Y_k\|_2^2}{n} + \lambda\|v\|_2^2. \tag{27}$$

Note that the gradient $\nabla L(v)$ of the regularized loss L is given by

$$\begin{aligned}\nabla L(v)/2 &= \sum_k S^\top X_k^\top (X_k S v - Y_k)/n + \lambda v = \sum_k S^\top M_k S v - \sum_k S^\top X_k^\top Y_k/n + \eta \\ &= Hv - \sum_k S^\top X_k^\top Y_k/n,\end{aligned}$$

where $H := S^\top M S + \lambda I_m \in \mathbb{R}^{m\times m}$, with $M := M_1 + M_2$ and $M_k := X_k^\top X_k/n$. Thus, setting $R := H^{-1}$, we may write

$$\widehat{v} = RS^\top (X_1^\top Y_1 + X_2^\top Y_2)/n = RS^\top (M_1 w_1 + M_2 w_2) + RS^\top X_1^\top E_1/n + RS^\top X_2^\top E_2/n.$$

Now, one deduces the bias-variance decomposition

$$E_{test}(\widehat{f}_{RP}) = \mathbb{E}_{\mathcal{D}}\mathbb{E}_{x\sim N(0,\Sigma_k)}[(\widehat{f}_{RP}(x) - x^\top w_1^*)^2] = \mathbb{E}_{X_1,E_1,X_2,E_2}\|S\widehat{v} - w_k\|_{\Sigma_k}^2 = B_k + V_k,$$

$$\begin{aligned}&\text{where } V_k := V_{k,1} + V_{k,2}, \text{ with } V_{k,j} := \frac{\sigma_j^2}{n}\mathbb{E}_{X_1,X_2} \operatorname{tr} S^\top M_j SRS^\top \Sigma_k SRS^\top, \\ &B_k := \mathbb{E}_{X_1,X_2}\|SRS^\top (M_1 w_1 + M_2 w_2) - w_k\|_{\Sigma_k}^2.\end{aligned}$$

The variance terms $V_{k,j}$ can be directly handled via FPT computations. We now look at the bias term B_k. We first treat the case $k = 1$. One has

$$\begin{aligned}&\mathbb{E}\|SRS^\top (M_1 w_1 + M_2 w_2) - w_1\|_\Sigma^2 \\ &= \mathbb{E}\|(SRS^\top (M_1 + M_2) - I_d)w_1 + SRS^\top M_2 \delta\|_\Sigma^2 \\ &= \mathbb{E}\|(SRS^\top M - I_d)w_1\|_\Sigma^2 + \mathbb{E}\|SRS^\top M_2 \delta\|_\Sigma^2 \\ &= \mathbb{E} \operatorname{tr} \Gamma(SRS^\top M - I_d)\Sigma(MSRS^\top - I_d) + \mathbb{E} \operatorname{tr} \Delta M_2 SRS^\top \Sigma SRS^\top M_2 \\ &= \operatorname{tr} \Gamma\Sigma + \operatorname{tr} \Gamma SRS^\top M\Sigma MSRS^\top - 2\mathbb{E} \operatorname{tr} \Gamma\Sigma SRS^\top M + \mathbb{E} \operatorname{tr} \Delta M_2 SRS^\top \Sigma SRS^\top M_2 \\ &= \underbrace{\operatorname{tr} \Gamma\Sigma + \mathbb{E} \operatorname{tr} \Sigma MSRS^\top \Gamma SRS^\top M - 2\mathbb{E} \operatorname{tr} \Gamma\Sigma SRS^\top M}_{\text{classical term } (B)} + \underbrace{\mathbb{E} \operatorname{tr} \Delta M_2 SRS^\top \Sigma SRS^\top M_2}_{\text{extra term } (\zeta)},\end{aligned}$$

where we recall that $R := (S^\top MS + \lambda I_m)^{-1}$ and $M := M_1 + M_2$ with $M_k = X_k^\top X_k$.

For the purposes of FPT computations, it might help to observe that $M_k = \lambda\Sigma_k^{1/2} Z_k^\top Z_k \Sigma_k^{1/2}$, where $Z_k := X_k \Sigma_k^{1/2}/(n\lambda)$ is an $n_k \times d$ random matrix with iid entries from $N(0.1/(n\lambda))$. Thus,

$$M_k = \lambda \overline{M}_k, \tag{28}$$

$$\overline{M}_k = \Sigma_k^{1/2} Z_k^\top Z_k \Sigma_k^{1/2}, \tag{29}$$

$$M = \lambda \overline{M}, \tag{30}$$

$$\overline{M} = \overline{M}_1 + \overline{M}_2 = \Sigma_1^{1/2} Z_1^\top Z_1 \Sigma_1^{1/2} + \Sigma_2^{1/2} Z_2^\top Z_2 \Sigma_2^{1/2}), \tag{31}$$

$$R = \overline{R}/\lambda, \tag{32}$$

$$\overline{R} = (S^\top \overline{M} S + I_m)^{-1} = \left(S^\top \Sigma_1^{1/2} Z_1^\top Z_1 \Sigma_1^{1/2} S + S^\top \Sigma_2^{1/2} Z_2^\top Z_2 \Sigma_2^{1/2} S + I_m\right)^{-1}. \tag{33}$$

We need minimal linear pencils for the random matrices

$$A\overline{M}_1 S\overline{R}S^\top BS\overline{R}S^\top, \tag{34}$$

$$A\overline{M}S\overline{R}S^\top BS\overline{R}S^\top \overline{M} \tag{35}$$

$$AS\overline{R}S^\top \overline{M}, \tag{36}$$

$$A\overline{M}_2 S\overline{R}S^\top BS\overline{R}S^\top \overline{M}_2, \tag{37}$$

in terms of the set of free variables $\{A, B, \Sigma_1^{1/2}, \Sigma_2^{1/2}, S, Z_1, Z_2, S^\top, Z_1^\top, Z_2^\top\}$. Observe that

$$\begin{aligned}&\operatorname{tr} AMSRS^\top BSRS^\top M \\ &= \operatorname{tr} AM_1 SRS^\top BSRS^\top M_1 + \operatorname{tr} AM_2 SRS^\top BSRS^\top M_2 + 2 \operatorname{tr} AMSRS^\top BSRS^\top M, \\ &\operatorname{tr} ASRS^\top M = \operatorname{tr} ASRS^\top M_1 + \operatorname{tr} ASRS^\top M_2.\end{aligned}$$